Alberto Saldana-Lopez

Postdoctoral Researcher · Space Telescope Science Institute (STScI)

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EXPERTISE

Determining how massive stars and active galactic nuclei (AGN) shaped cosmic reionization, using space-based (HST, JWST) and ground-based (Keck, VLT) far-ultraviolet to infrared imaging and spectroscopy. Following ionizing photons from the sources that produce them, through the gas and dust of the interstellar medium, to the circumgalactic medium where stellar feedback deposits its energy, to understand the physical processes that regulate ionizing photon escape.

SUMMARY

- **Publications:** 69 papers (**9 first/second author**, 11 with major contribution); 2,984 citations; **h-index 30**; first-author papers alone **>390 citations** (**SciX**, as of October 2026).
- **Observing programs:** 4 PI programs: HST (63 orbits), JWST (9.4 h), VLT (32 h); Co-I on 26 further HST, JWST, VLT, Keck and GMRT programs.
- **Grants:** PI: $274,647 (HST Cycles 33 and 34; JWST Cycle 5).
- **Software:** developer of **FiCUS**, a public Python code for fitting the stellar continuum of far-UV galaxy spectra, adopted in 10 independent studies.
- **Mentoring:** co-advisor of 1 Ph.D. student and advisor of 2 B.Sc. theses; guest lecturer in 2 graduate courses.
- **Talks:** 10 invited talks, 20 contributed talks.
- **Collaborations:** key member of 7 international collaborations.
- **Scientific leadership in collaborations:** co-lead of the Lyα image processing and MAST data release of LaCOS (GO-17069); lead of the Keck/KCWI emission-line analysis of HIGHLIGHTS (GO-18402); core member of the LzLCS (GO-15626). My LaCOS results directly motivated the design of the *HyperDeep Ultraviolet Field* (GO-18004, 124 HST orbits; Co-I Saldana-Lopez), set to become the deepest UV image of the sky.

EMPLOYMENT / APPOINTMENTS

2026 – present **Postdoctoral Researcher, Science Mission Office**
Space Telescope Science Institute (STScI), Baltimore, USA
Advisors: Dr. Alaina Henry, Dr. Bethan James, Dr. Travis Fischer

2023 – 2026 **Postdoctoral Fellow, Galaxies Group**
Department of Astronomy, Stockholm University, Sweden
Advisor: Prof. Matthew J. Hayes

2019 – 2023 **Doctoral researcher and Teaching Assistant**
Department of Astronomy, University of Geneva, Switzerland
(TA for the undergraduate Astrophysics Lab, 2019 – 2021)

2018 – 2019 **Research Assistant, High Energy Group**
Complutense University of Madrid (UCM), Spain

2017 – 2018 **Research Assistant, Nanophotonics Group**
International Iberian Nanotechnology Laboratory (INL), Portugal

2016 – 2017 **Research Assistant, Dark Energy Survey Group**
CIEMAT, Madrid, Spain

2015 – 2016 **Research Assistant, Laser Processing Group**
CSIC, Madrid, Spain

EDUCATION

2019 – 2023 **Ph.D. in Astronomy and Astrophysics, University of Geneva, Switzerland**
Thesis: *Properties of star-forming galaxies contributing to cosmic reionization*
Advisor: Prof. Daniel O. Schaerer · Ph.D. awarded in September 2023

2018 – 2019 **M.Sc. in Astrophysics, Complutense University of Madrid, Spain**
Thesis: *An observational determination of the evolving extragalactic background light from the multiwavelength HST/CANDELS survey in the Fermi and CTA era*
Advisors: Dr. Alberto Domínguez, Dr. Pablo G. Pérez-González

2014 – 2018 **B.Sc. in Physics, Complutense University of Madrid, Spain**

GRANTS AND AWARDS

2026 **$65,999** — HST GO-18352, Cycle 34 research grant (PI)

2026 **$87,248** — JWST GO-12405, Cycle 5 research grant (PI)

2025 **$121,400** — HST GO-18034, Cycle 33 research grant (PI)

2026 **$1,000** — Alva and Lennart Dahlmark research grant

2026 **$1,500** — STScI Spring Symposium travel grant

2025 **$1,500** — STScI Spring Symposium travel grant

2025 **$1,500** — Royal Swedish Academy of Sciences (KVA) mobility grant

2024 **$1,800** — Simons Foundation (Flatiron Institute, CfA) travel grant

2023 **$1,500** — Swiss Society for Astronomy and Astrophysics (SSAA) mobility grant

2021 **Ramón Corbalán Prize** for the popularization and education in Nonlinear and Quantum Optics, for “Two-photon polymerization”, **Saldana-Lopez et al.** (2020), The Spanish Journal of Physics, 34, 2

OBSERVING PROGRAMS

PI of 4 and Co-I of 26 accepted programs with space- and ground-based facilities. The three HST and JWST PI programs provide the core data of my proposed NHFP research program.

As Principal Investigator (4)

HST Cycle 34 — **Constraining the ionizing and FUV output of Little Red Dots with HST: a decisive test of their role in Cosmic Reionization**
PID 18352 · HST/COS · 23 orbits · $65,999 research grant

JWST Cycle 5 — **Illuminating the origins of extreme OVI emission in the Circumgalactic Medium of J1156 with JWST/MIRI**
PID 12405 · JWST/MIRI · 9.4 hours · $87,248 research grant

HST Cycle 33 — **Spatially resolving the conditions for ionizing radiation escape in galaxies**
PID 18034 · HST/ACS · 40 orbits · $121,400 research grant · featured in the STScI Newsletter

ESO P112 — **The nature of UV emission line galaxies: a study of CIV emitters at Cosmic Noon**
ID 112.2639 · VLT/X-Shooter · 32 hours

As Co-Investigator (26)

HST Cycle 34 — **Lyman Continuum and Lyman Alpha escape in the high-ionization conditions observed in the Epoch of Reionization**
PI: Chisholm · PID 18435 · COS · 34 orbits

HST Cycle 34 — **Illuminating the Physics of the Baryon Cycle with HIGHLIGHTS: the HST Imaging of Galaxy Halo LIGHT Survey**
PI: Rubin · PID 18402 · ACS/WFC3 · 160 orbits

HST Cycle 34 — **A deep dive into the interstellar forges shaping Lyman alpha emission**
PI: Le Reste · PID 18361 · COS · 29 orbits

HST Cycle 33 — **The HyperDeep Ultraviolet Field**
PI: Hayes · PID 18004 · WFC3 · 124 orbits

HST Cycle 33 — **Unlocking the full potential of JWST spectroscopic fields with SHIP3: Snapshot HST Imaging of Pure-Parallel Programs**
PI: Nedkova · PID 18022 · WFC3 · 123 orbits

HST Cycle 33 — **The High Redshift Lyman Continuum Survey**
PI: Scarlata · PID 18080 · WFC3 · 53 orbits

Keck 2025A — **Multiphase Galactic Outflows with HIGHLIGHTS**
PI: Coil · ID U240 · KCWI · 10 hours

HST 32 Bridge — **Bridge to the Frontiers of Time Domain: Supermassive Black Holes and Exotic Stellar Transients in the Early Universe**
PI: Hayes · PID 17908 · WFC3 · 20 orbits

JWST DDT	**Let there be Light: Directly Witnessing the Birth of Metal-Free, Pop III Stars in an Ultra-Faint Galaxy at z = 6.5** PIs: Fujimoto & Naidu · PID 9223 · NIRSpec · 39 hours
JWST Cycle 4	**Formation and nature of the UV-brightest starbursts in the distant Universe** PI: Marques-Chaves · PID 8258 · NIRSpec · 38 hours
HST Cycle 32	**MgII maps to reveal how ionizing photons escape local LyC emitting galaxies** PI: Leclercq · PID 17761 · ACS · 31 orbits
HST Cycle 32	**Lyman alpha imaging of galaxies with the lowest mass and metallicity** PI: Östlin · PID 17826 · ACS/WFC3 · 48 orbits
JWST Cycle 3	**Ionization and Obscuration in LyC Emitters: A MIR Look at Lyman Continuum Escape** PI: Flury · PID 5554 · MIRI · 31 hours
JWST Cycle 2	**JWST's GLIMPSE: gravitational lensing & NIRCam imaging to probe early galaxy formation and sources of reionization** PI: Atek · PID 3293 · NIRCam · 155 hours
JWST Cycle 1	**The First Observations of the Ionizing Luminosity of Galaxies within the Epoch of Reionization** PI: Chisholm · PID 1871 · NIRSpec · 22 hours
JWST Cycle 1	**LyC22 – Deep spectroscopic insights on star-forming galaxies 2.2 Gyr after the Big Bang** PI: Schaerer · PID 1869 · NIRSpec · 73 hours
HST Cycle 31	**Establishing the Geometry of Lyman Continuum Escape** PI: Carr · PID 17443 · COS · 23 orbits
HST Cycle 31	**High-resolution imaging of the ionizing and non-ionizing radiation of extreme starbursts at z ~ 2.4** PI: Marques-Chaves · PID 17424 · WFC3/UVIS · 23 orbits
HST Cycle 30	**Far-Ultraviolet Legacy Survey of the GOODS and COSMOS Fields: Completing the Census of the UV Sky** PI: Siana · PID 17032 · ACS/SBC · archival
HST Cycle 30	**The Lyman-alpha and Continuum Origins Survey (LaCOS)** PI: Hayes · PID 17069 · ACS, WFC3 · 119 orbits
HST Cycle 30	**Resolving Lyman Alpha emission in a complete sample of Lyman Continuum leakers and non-leakers** PI: Leclercq · PID 17153 · COS · 49 orbits
HST Cycle 30	**Revealing the link between strong LyC emitters and enigmatic CIV emitters** PI: Schaerer · PID 17169 · STIS · 34 orbits
GMRT	**Radio-SED Study of low-z Lyman Continuum Emitters** PI: Bait · GMRT Bands 3/4/5 · 24 hours

ESO P109 **Observations of the ionizing spectra in the Lyman continuum of distant starbursts**
PI: Marques-Chaves · ID 109.23G1 · VLT/FORS2 · 15 hours

ESO P108 **XShooter Survey of Extremely UV and Lyα Luminous Star-forming Galaxies at z = 2–3.6**
PI: Marques-Chaves · ID 108.228N · VLT/X-Shooter · 18 hours

ESO P106 **Deep spectroscopy of low-z HST Lyman continuum emitters: revealing their ISM and ionizing radiation field properties**
PI: Schaerer · ID 106.215K · VLT/X-Shooter · 15 hours

TEACHING AND MENTORING

2023 – present **Ph.D. thesis co-advisor: Alice R. Young, Stockholm University**
Observational constraints on Supermassive Black Hole seeding mechanisms
First-author paper: Young et al. (2026), "Glimmers in the Cosmic Dawn. III. On the Photometrically Determined Black Hole Mass to Stellar Mass Relation Across Cosmic Time", ApJ, 999, 175; one more paper in preparation.
Her work, together with Hayes et al. (2024, Glimmers I), was featured in press releases by Stockholm University (Sep 2024) and Advanced Science News (Oct 2024)

2025 – 2026 **B.Sc. thesis advisor: Serien Ali, Stockholm University**
Are Variable Supermassive Black Holes the Engines of Little Red Dots?

2024 – 2025 **B.Sc. thesis advisor: Pontus Strand, Stockholm University**
An unbiased sample of strong UV emitters at Cosmic Noon

2024 – 2025 **Ph.D. thesis committee member: Olof Nebrin, Stockholm University**
Lyα feedback prevails at Cosmic Dawn

2024 – 2025 **Guest lecturer, Observational Astrophysics I (AS7003, graduate), Stockholm University**
Guided discussions on high-redshift galaxies with JWST

2023 – 2024 **Guest lecturer, Physics of the Interstellar Medium (AS7001, graduate), Stockholm University**
Led a practical exercise on dust attenuation in galaxies

2019 – 2021 **Teaching Assistant, Astrophysics Lab (13P950, undergraduate), University of Geneva** Design of Python workflows to deal with galaxy catalogs (HST, SDSS)

SERVICE

Conferences **2026 STScI Spring Symposium: The Metal-Poor Frontier**
Chair of Sessions S9 and S10 ('Feedback'), and instructor in Session 19 ('Modeling Tools'): FiCUS code tutorial

Organization	**Organizer, OKC Galaxy Formation and Evolution Journal Club** Department of Astronomy, Stockholm University · 2025 – 2026
Collaborations	**Data-analysis leadership in HST surveys** Co-lead of the Lyα image processing and MAST data release for LaCOS; lead of the Keck/KCWI and KCRM emission-line analysis for HIGHLIGHTS
Refereeing	**Referee for ApJ, MNRAS and A&A** 4 manuscripts reviewed for The Astrophysical Journal (ApJ), Monthly Notices of the Royal Astronomical Society (MNRAS), and Astronomy & Astrophysics (A&A)

SOFTWARE AND DATA PRODUCTS

FiCUS — FItting the stellar Continuum of Uv Spectra · github.com/asalda/FiCUS

Lead developer and maintainer of this public Python code, which fits the stellar continuum of far-UV galaxy spectra with stellar population synthesis models and dust attenuation, delivering stellar ages, metallicities, UV attenuation and ionizing photon production rates. Designed for HST/COS spectra of the Low-Redshift Lyman Continuum Survey (**Saldana-Lopez et al. 2022**) and applied to deep rest-frame UV spectra at $3 \leq z \leq 5$ (VANDELS; **Saldana-Lopez et al. 2023**); now used by the community in 10 independent papers to model the stellar populations of local and high-redshift galaxies, and taught at the 2026 STScI Spring Symposium ('Modeling Tools' session).

HIGHLIGHTS Keck/KCWI emission-line pipeline

Lead developer of the emission-line fitting pipeline for the Keck/KCWI and KCRM data cubes of the HIGHLIGHTS galaxies, used to map ionized outflows around nearby starbursts.

High-Level Science Products for MAST

Processing of HST ACS/SBC, ACS/WFC and WFC3/UVIS imaging into science-ready Lyα and continuum maps for LaCOS, and contribution to the preparation of the corresponding High-Level Science Products for the Mikulski Archive for Space Telescopes (MAST).

TALKS AND SEMINARS

Invited seminars and colloquia (10)

Nov 2025	**Extended Lyα halos as probes of ionizing radiation escape near and far** FCAD Astronomy Seminar, University of Massachusetts Amherst, USA
Oct 2025	**A low-redshift look to reionization with star-forming galaxies** Astronomy Seminar, University of Minnesota, USA
Oct 2024	**The average UV emission line spectra of high-redshift galaxies** STScI/JHU Galaxies Journal Club, STScI, Baltimore, USA

Nov 2023 **The properties of star-forming galaxies contributing to reionization**
Astronomy Seminar, Stockholm University, Sweden

May 2023 **A low-redshift look to reionization with star-forming galaxies**
Colloquium, Herzberg Astronomy & Astrophysics Research Centre (NRC), Canada

Nov 2022 **Ionizing properties of galaxies through the eyes of absorption spectroscopy**
Lunch Seminar, University of Texas at Austin, USA

Mar 2021 **An observational determination of the Extragalactic Background Light**
AstroSeminar, Caltech/IPAC, USA (online)

Feb 2021 **An observational determination of the Extragalactic Background Light**
Astronomy Seminar, University of Geneva, Switzerland (online)

Jan 2021 **An observational determination of the Extragalactic Background Light**
Astronomy Seminar, University of California Riverside, USA (online)

Jan 2021 **An observational determination of the Extragalactic Background Light**
Cosmo Club Seminar, University of California Santa Cruz, USA (online)

Contributed conference talks (20)

May 2026 **Extended Lyα halos as probes of ionizing radiation escape**
STScI Spring Symposium, Baltimore, USA

Apr 2026 **A glimpse to outflows in the lowest-mass galaxies**
Charting Cosmic Dawn, Copenhagen, Denmark

May 2025 **Feedback and dynamical masses in high-z galaxies**
Cosmic Frontier Center 2025 Conference, Austin, USA

May 2025 **Extended Lyα halos as probes of ionizing radiation escape**
STScI Spring Symposium, Baltimore, USA

Apr 2025 **Feedback and dynamical masses in high-z galaxies**
First Galaxies Meeting, Oxford, UK

Jul 2024 **A low-redshift look to reionization with star-forming galaxies**
Cosmic Dawn at High Latitudes Workshop, Stockholm, Sweden

May 2024 **The Lyman-alpha and Continuum Origins Survey**
First Stars VII, New York City, USA

Jan 2024 **Ionizing properties of galaxies through the eyes of absorption spectroscopy**
Linking Galaxy Physics from ISM to IGM Scales, Sexten, Italy

Jan 2024 **Constraining galactic feedback at Cosmic Dawn**
DLOCKS-24 Workshop on Galaxy Evolution, Copenhagen, Denmark

Apr 2023 **Ionizing properties of galaxies through the eyes of absorption spectroscopy**
Escape of Lyman Radiation from Galactic Labyrinths, Crete, Greece

Sep 2022 **An observational determination of the Extragalactic Background Light**
CRPropa Workshop on Astroparticle Propagation, Madrid, Spain

Jul 2022 **The ionizing properties of star-forming galaxies at 3 ≤ z ≤ 5**
From Galaxies to Cosmology with Spectroscopic Surveys, Marseille, France

Jun 2022 **The ISM properties of low-z Lyman Continuum emitters**
European Astronomical Society (EAS) Annual Meeting, Valencia, Spain

Mar 2022 **The ISM properties of low-z Lyman Continuum emitters**
The Growth of Galaxies in the Early Universe VII, Sexten, Italy

Jan 2022 **Unveiling the ISM properties of low-z Lyman Continuum emitters**
Production and Escape of Lyman Photons through Time and Space, UK (online)

Nov 2021 **Unveiling the ISM properties of low-z Lyman Continuum emitters**
SAZERAC-SIP: Early Galaxy Formation Near and Far (online)

May 2021 **Unveiling the ISM properties of low-z Lyman Continuum emitters**
STScI Workshop: MOS for Measures of Galaxy Evolution (online)

Apr 2021 **An observational determination of the Extragalactic Background Light**
Ninth International Fermi Symposium (online)

Apr 2021 **Using LIS UV-lines to select Lyman continuum leaking candidates**
Extragalactic Spectroscopic Surveys: Past, Present and Future (online)

Sep 2019 **An observational determination of the Extragalactic Background Light**
VII Meeting on Fundamental Cosmology, Madrid, Spain

OUTREACH

Feb 2025 **AstroBites feature: "Fueling or Starving? The Role of Gas Flows in Early Galaxy Evolution"** L. Rowland, based on **Saldana-Lopez et al.** (2025, MNRAS, 544, 132)

Sep 2022 **Newspaper interview: "Reencuentro galáctico en Canarias"** J. Dávila, El Día (La Opinión de Tenerife)

Sep 2020 **Outreach article: "Two-photon polymerization" Saldana-Lopez et al.** (2020), The Spanish Journal of Physics, 34, 2 – Awarded the Ramón Corbalán Prize 2021 for the popularization and education in Nonlinear and Quantum Optics

Dec 2019 **"Extraterrestrial Life"** Public outreach talks at high school level

2018 – 2020 **Contributor, *El Blog de Laniakea* (popular science blog in Spanish)**
elblogdelaniakea.wordpress.com

CONTINUED DEVELOPMENT

May 2022 **Severo Ochoa Advanced School on Galaxy Evolution**
IAA-CSIC, Granada, Spain

Jul 2021 **International Summer School on the ISM of Galaxies**
CNRS, France (online)

Jun 2021 **Summer School in Statistics for Astronomers XVI**
Penn State University, USA (online)